MILITARY SPECIFICATION

CONDUIT: ELECTRICAL, FLEXIBLE, SHIELDED, ALUMINUM ALLOY FOR AIRCRAFT INSTALLATIONS

This amendment forms a part of Military Specification MIL-C-6136A, 10 December 1965, and is mandatory for use by all Departments and Agencies of the Department of Defense.

Page 1

2.1 Specifications - Military· Delete reference to:

"MIL-R-6855, Rubber, Synthetic, Sheeted, Molded and Extruded" and substitute "MIL-S-6855, Synthetic Rubber Sheets, Strips, Molded or Extruded Shapes".

Page 2

Add the following paragraph:

"2.2 Other publications - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials

ASTM D 149-64	Dielectric Breakdown Voltage and Dielectric Strength of Electrical Insulating Materials at Commercial Power Frequencies
ASTM D 412-68	Tension Testing of Vulcanized Rubber
ASTM D 568-61	Flammability of Plastics 0.127 CM (0.050 inch) and Under in Thickness
ASTM D 570-63	Water Absorption of Plastics
ASTM D 624(c)-54	Test for Tear Resistance of Vulcanized Rubber

FSC 5975

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T06:39Z
Check the source to verify that this is the current version before use.

American Society for Testing and Materials (Continued)

ASTM D 746-64T	Test for Brittleness Temperature of Plastics and Elastomers by Impact
ASTM D 792-66	Test for Specific Gravity and Density of Plastics by Displacement
ASTM D 2240-64T	Test for Indentation Hardness of Rubber and Plastics by means of a Durometer

(Copies of ASTM publications may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103)"

3.1.4. Delete in its entirety and substitute the following:

"3 1 4 Outer covering material (Type II) - The outer cover of Type II conduit shall be either synthetic rubber or nitrile rubber plasticized polyvinyl chloride or nitrile rubber plasticized polyvinyl chloride - acetate

3.1 4 1 Synthetic rubber - The synthetic rubber shall conform to Class 2 of MIL-S-6855 and shall be of uniform thickness and free from pits and pinholes.

3 1 4.2 Nitrile rubber plasticized polyvinyl chloride or nitrile rubber plasticized polyvinyl chloride - acetate - The nitrile rubber plasticized polyvinyl chloride or nitrile rubber plasticized polyvinyl chloride-acetate shall be of uniform thickness and free from pitting and shall conform to Table I "

TABLE I

PROPERTY	VALUE	APPLICABLE ASTM OR FEDERAL TEST PROCEDURE
Tensile strength (psi, min.)	1800	ASTM D 412
Elongation (%) Min	350	ASTM D 412
Modulus at 100% elongation (psi)	900	ASTM D 412
Tear strength (lbs/inch) Min.	350	ASTM D 624 Die (c)
Water absorption (gram/sq in)	04 max	ASTM D 624
Hardness (Durometer ±3)	82	ASTM D 2240

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T06:39Z
Check the source to verify that this is the current version before use.

TABLE I (Continued)

PROPERTY	VALUE	APPLICABLE ASTM OR FEDERAL TEST PROCEDURE
Specific gravity	1.20 max.	ASTM D 792
Low temperature flexibility(° F)	-70	Fed. Test Method Std. No. 601, Method 5211
Flammability	Self-extinguishing	ASTM D 568
Dielectric strength (volts/mil) Minimum	600	ASTM D 149 (Short time in air-0.30 mil sheet) 2-inch diameter electrodes

3.2.1.1, 3.2.1.2, and 3.2.2.1: Delete reference to "Table I" and substitute "Table II".

Page 4

Change existing Table I to Table II.

Change existing Table II to Table III.

Page 5

3.2.2.2, 3.2.2.3, 3.2.3.1, 3.2.3.2, and 3.4: Delete references to "Table I" and substitute "Table II".

3.2.4: Delete reference to "MIL-R-6855" and substitute "MIL-S-6855".

3.4, lines 5 and 6: Delete reference to "Table II" and substitute "Table III".

Page 7

4.5.1 and 4.5.3: Delete reference to "Table I" and substitute "Table II".

Page 8

Figure 2: Delete reference to "Table I" and substitute "Table II".

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T06:39Z
Check the source to verify that this is the current version before use.

6.2(c): Delete reference to "Table I" and substitute "Table II".

Custodians:
Army - EL
Navy - AS
Air Force - 85

Review activities:
Army - El
Navy - AS
Air Force - 85

Preparing activity:
Navy - AS
(Project No. 5975-0404)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T06:39Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS. In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 12503 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commander
Naval Air Systems Command (AIR 51122)
Washington, DC 20361

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T06:39Z
Check the source to verify that this is the current version before use.